# AI-Powered IT Project Risk Management Using a Multi-Agent Intelligence Framework

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***Abstract*—The AI-Powered Project Risk Management System is a real-time platform that identifies and mitigates risks in IT projects using advanced AI and machine learning. It integrates multi-agent orchestration (CrewAI), risk scoring (Scikit-learn), sentiment analysis (HuggingFace), and semantic search (ChromaDB), delivered through a Streamlit dashboard with a FastAPI backend. Key features include automated risk classification, a multi-agent workflow, contextual insights from historical data, and a chatbot interface for natural language queries. The system improves early risk detection, project visibility, and decision-making in dynamic IT environments.**

***Index Terms*—Risk management, CrewAI, Scikit-learn, Huggingface, chatbot, classification**

## I. Introduction

### A. Problem Statement

IT projects work in environments where things like deadlines change people don't have resources the quality isn't right money goes over budget and outside events keep happening all through the project. Catching these problems early is important because if you don't spot them quickly you don't have time to fix them.. Traditional ways of managing risk in projects usually depend on checking things at regular times and having people write reports by hand. This makes it hard to keep up with the changes that happen all the time during a project.

There are problems with this method. First finding risks can happen after the problem starts to affect the project. Second if you check once in a while you might miss what is changing between those checks. Third the lessons from projects don't always get used when looking at new ones. Finally details like how on schedule the project is, how much people are working how good the quality is and what is happening outside the project can be, in places. This makes it hard to get a picture.

These problems push for a way to handle project risks. The new system should be able to look at all kinds of project data and give clear risk reports and advice.

### B. Motivation & Research Gaps

The growing availability of organized project data and unorganized project details creates a chance to automate some parts of the risk-evaluation process. Machine-learning methods can be used to find trends in data while natural language processing can pull out clues from written information. Also semantic search can help get information from past project situations.

These abilities are usually looked at on their own. A system for managing risk might use data without adding written information or it might use language-based analysis without linking it to numbers about the project. Also just using analysis methods doesn't give a way to bring their results together and make one overall risk evaluation.

This research therefore looks at an approach where different AI agents work together on various parts of project-risk analysis. The suggested system brings together organized project details, written clues, outside context and past knowledge, within one shared process.

### C. Objectives

The main goal of this work is to create a system that helps with checking risks, in IT projects all the time. The system is meant to:

- Look at project-health data that is organized and also check text that is not organized to find risks;
- Use machine learning and language processing methods to find different signs of risk;
- Work together on different tasks using a team of agents;
- Bring in past information using a way to find meaning in the data;
- Make clear risk reports and suggest ways to deal with the risks through a tool that people can use easily.

### D. Scope of Study

We see that the proposed system focuses on risk assessment in IT projects looking at both project-health indicators and external contextual information. Internal indicators include schedule, budget, resource utilization and project performance. External information can give context about factors that may influence risk assessment.

We evaluate the implementation using project data that is available inside the prototype environment. We plan to integrate the system with project-management and development platforms such as Jira and GitHub, in future work. We intend the system to act as a decision-support framework. The risk. Recommendations that the system generates are not meant to replace human judgment by project managers.

## II. Related Work

### A. Machine Learning for Software Project Risk Prediction

Traditional software project risk management has relied heavily on expert judgment, predefined risk categories, and periodic assessment. With the increasing availability of project datasets, researchers have investigated machine-learning techniques for identifying and predicting software-related risks. Khan et al. studied tree-family machine-learning techniques for predicting risks associated with software requirements and evaluated multiple models using standard classification metrics [1]. Their work demonstrates the applicability of supervised learning to early risk prediction, particularly when structured risk-related datasets are available. Similarly, systematic studies on machine-learning-based prediction in software engineering have explored the use of predictive models for identifying failure-prone software components and supporting project decision-making [2].

Research has also examined risk management in agile development environments. Shehzad et al. reviewed existing approaches to risk management in agile software development and identified areas including risk analysis, scalability risks, adoption risks, and risk-management frameworks [3]. These studies indicate that machine learning and structured analytical approaches can support risk identification; however, such approaches generally focus on specific risk categories or structured datasets rather than combining multiple heterogeneous sources of project information.

### B. Natural Language Processing for Risk Identification

Project-related risk information is not limited to numerical indicators. Important signals can also appear in reports, communications, news articles, requirements, and other textual sources. Consequently, natural language processing (NLP) has increasingly been investigated for automated risk identification and classification.

For example, research on context-aware risk identification has demonstrated the use of NLP to extract risk-related information from textual documents. A multi-sentence context-aware approach was proposed to identify project-level risk information from news articles, showing that contextual information across multiple sentences can improve automated risk classification [4]. Other research has combined NLP and machine-learning techniques to classify risk-related statements in engineering documents, demonstrating the potential of automated text analysis for reducing the effort required for manual risk identification [5].

NLP has also been used to provide decision-support interfaces for risk management. Chou et al. developed a deep-learning-based chatbot for risk management in river-dredging projects, combining a project-risk knowledge base with NLP-based text matching to provide risk information and response recommendations to users [6]. While this work demonstrates the value of combining textual analysis, knowledge bases, and interactive interfaces, its application is domain-specific and does not integrate textual risk signals with broader software project-health metrics.

### C. Multi-Agent Approaches for Software Project Management

Another research direction involves the use of multi-agent systems to distribute complex project-management activities among specialized computational agents. Baia proposed an integrated multi-agent-based simulation approach for software project management that incorporated project knowledge, development processes, situation-awareness information, and learning techniques to support project decision-making [7]. The work highlighted the potential of multi-agent approaches for handling dynamic project conditions and supporting more proactive management.

More recently, multi-agent architectures have also been explored for software engineering activities. For example, recent research has investigated domain-specific AI agents for automating different stages of the software development lifecycle, demonstrating how specialized agents can coordinate tasks such as requirements analysis, planning, and architectural modeling [8]. These developments suggest that agent-based architectures can provide a flexible mechanism for decomposing complex software-engineering workflows into specialized tasks.

However, existing multi-agent approaches do not necessarily address the specific problem of continuously combining quantitative project-health indicators, textual risk signals, external context, and historical project knowledge into a unified risk assessment. This motivates our use of specialized agents for different stages of project-risk analysis rather than treating the complete risk-management workflow as a single analytical task.

### D. Historical Knowledge and Semantic Retrieval for Risk Management

Historical project information can provide valuable context when evaluating newly emerging risks. Instead of treating every risk situation independently, retrieval-based approaches can allow previously observed situations and corresponding mitigation strategies to be reused.

Research in other project-risk domains has demonstrated the usefulness of knowledge-based retrieval for risk decision support. Chou et al., for instance, developed a risk knowledge base that allowed users to retrieve relevant risk-prevention information and response measures through natural-language interaction [6]. Similarly, recent knowledge-augmented approaches have investigated the integration of domain-specific knowledge with language models for automated risk identification, particularly in domains where specialized knowledge is required [9].

These approaches indicate that historical and domain knowledge can complement predictive models and text-based analysis. Nevertheless, knowledge retrieval is often implemented as an independent information-access mechanism. In our framework, we incorporate semantic retrieval as one component of a broader risk-analysis pipeline, allowing historical information to provide contextual evidence alongside structured project metrics and textual signals.

### E. Research Gap

The reviewed literature demonstrates that several important components of automated risk management have already been investigated independently. Machine-learning methods can support structured risk prediction [1], [2], while NLP techniques can extract risk-related information from unstructured text [4], [5]. Knowledge-based systems can provide historical risk information and recommendations [6], [9], and multi-agent approaches can distribute complex software-management activities among specialized components [7], [8].

However, we identify a gap in the integration of these capabilities into a unified IT project-risk management workflow. Existing studies generally emphasize one or a subset of these capabilities, such as structured risk prediction, textual risk identification, knowledge retrieval, or agent-based project management. Our work addresses this gap by developing a hybrid framework in which we combine structured project-health analysis, NLP-based contextual analysis, machine-learning-based risk scoring, historical semantic retrieval, and report generation within a coordinated multi-agent architecture.

Through this architecture, we aim to move from isolated risk-analysis techniques toward an integrated decision-support workflow. Rather than replacing individual analytical methods, our approach coordinates their outputs so that multiple sources of evidence can contribute to a consolidated project-risk assessment.

## III. Proposed Methodology

### A. Overview

We propose a mixed AI system for risk check in IT projects. Our method merges structured project health data with text and background data to produce a combined evaluation of project risk. Of depending on one prediction model we split the risk analysis into specific steps and coordinate them through a team of agents.

Our framework takes data from sources, such as project progress, task status, schedule details, resource signs, money data and outside background data. First the collected data is. Changed into forms that fit later risk analysis. Structured project signs are examined with machine learning while text is studied with text analysis. Past project data is fetched by looking for meaning similarity to give background during risk check.

The resulting analysis results are joined by the risk analysis flow to create a risk score and matching project risk category. Finally we create a report, with the identified risk factors supporting notes and suggested ways to reduce risk. We show these results on a dashboard and chat interface.

The overall methodology can therefore be represented as:

Data Collection → Data Preprocessing → Multi-Source Analysis → Risk Scoring → Historical Context Retrieval → Risk Report Generation → Visualization

### B. Data Collection

We look at two types of information in our framework: internal project information and external contextual information.

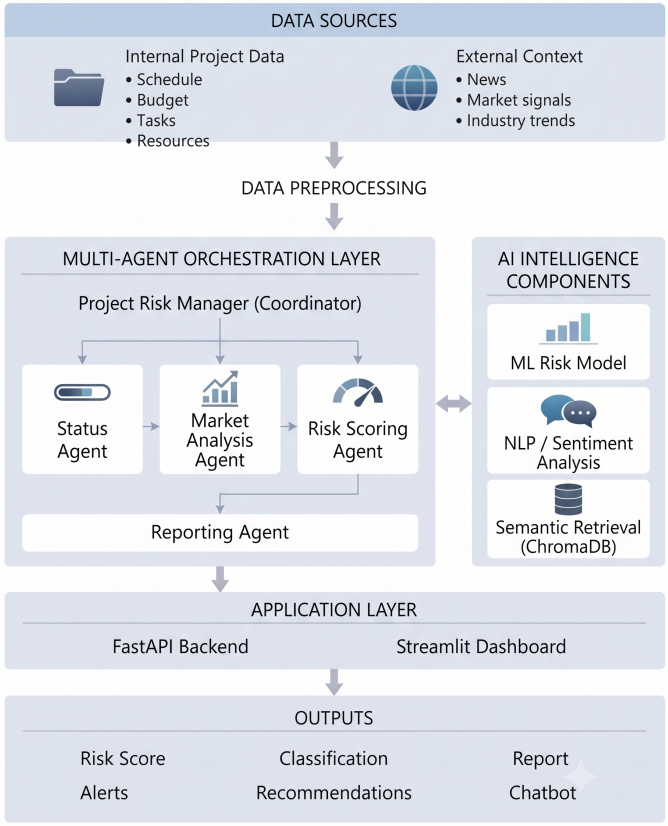



Fig. 1. Overall architecture of the proposed AI-powered IT project risk management framework

Internal project information reflects how an IT project is doing now. It can include things like:

- How much of the project has been completed, in terms of percentage
- The status of tasks and milestones
- Details about the project schedule
- Budget updates and financial metrics
- Information about team members and resources
- Data on development progress and quality
- Any issues that have been reported

External contextual information gives us insights that could affect the risk level of the project. This might involve:

- News about the markets or the industry
- Changes in the market
- Economic conditions
- New developments in technology
- Other outside events that could matter

Besides the current data we also keep historical project information. This data can be accessed later using similarity so we can compare what's happening now with past project risks.

Now our prototype works with data that's already inside the system. In the future we plan to connect it to live sources, like Jira and GitHub.

### C. Data Preprocessing

Before doing risk analysis we get the information ready so it is easier to use and more consistent in parts of the analysis.

For information that is organized in a way we do things like fix errors in the data deal with missing parts make sure numbers are in the same format and change words into a way that computers can understand.

For text we get it ready for processing by a computer. We take the parts of the text and turn them into a form that can be used to find feelings and possible risks in the content.

The step before analysis gives us two types of information:

Structured features, which show numbers and facts about the project.

Unstructured representations, which have information from text and other sources.

This split helps us use the methods, for each type of information while still using a similar way to look at the risk later on.

### D. Structured Risk Analysis

We use machine learning methods to examine organized project data and estimate how likely project risk conditions are.

Our present machine learning part has supervised classification models. In the prototype we train Logistic Regression and Random Forest classifiers on risk features. We split data 80 percent for training and 20 percent for testing. We set a fixed random seed so results can be repeated.

The classification part gives predictions that can serve as evidence when assessing project risk. We measure the models with metrics such as accuracy, precision, recall, F1 score and confusion matrices.

The Random Forest model is then used as the predictive part, in the prototype risk scoring pipeline.

This method lets us turn raw project indicators into a machine-learning risk signal of depending only on manually set thresholds.

### E. Use of NLP

Structured project metrics alone may not capture all relevant risk information. We therefore incorporate NLP-based analysis to process textual information and identify additional signals.

Our framework uses sentiment analysis to examine the overall tone of relevant textual information. Negative or deteriorating sentiment can provide an additional indication of potential issues when interpreted alongside structured project metrics.

We also process external textual information, such as relevant financial or industry news, to identify contextual factors that may affect a project's risk profile.

The resulting NLP signals are not treated as independent final decisions. Instead, we use them as complementary evidence that can be considered together with structured project-health indicators.

This allows our framework to incorporate both quantitative project information and qualitative textual information into the risk-analysis workflow.

### F. Historical Knowledge Retrieval

We incorporate historical project information through semantic retrieval using ChromaDB.

Historical project records and risk-related information are stored as searchable representations. When a new project situation is analyzed, we generate a semantic representation of the current context and retrieve historically relevant information.

The retrieved information can provide:

Similar previously observed project situations Previously identified risk factors Historical project observations Relevant mitigation strategies

The purpose of this component is to provide additional context rather than directly determine the final risk classification. The retrieved knowledge is supplied to the subsequent analysis workflow so that our system can consider historical evidence alongside current project information.

### G. Risk Scoring and Classification

I think after each analytic stage finishes we combine the signals into the risk assessment workflow.

Our system gives a risk score from 0 to 100. That risk score is matched to project health categories: On Track, In Progress, At Risk

I think the risk assessment looks at evidence, from project indicators machine-learning predictions, text signals and past records.

Importantly the risk score is a tool to help decisions, not a decision. Project people can use the risk score to spot areas that need review or fixes.

The exact weights and math of the risk score must be written down using the code we built. We should not put a math formula in the paper until we check the real scoring code.

### H. Risk Report Generation

Following risk assessment, we generate a structured report that summarizes the current project condition.

Our reporting stage contains:

Overall risk assessment Identified risk factors Supporting project observations Relevant contextual information Risk score and classification Suggested mitigation actions

The generated report provides a human-readable representation of the outputs produced by the different analytical components.

## IV. Multi-Agent Architecture

We implement our risk-management workflow using a multi-agent architecture in which specialized agents perform different analytical tasks under the coordination of a central Project Risk Manager. This decomposition allows us to separate project-status analysis, external-context analysis, risk scoring, and report generation while maintaining a common workflow.

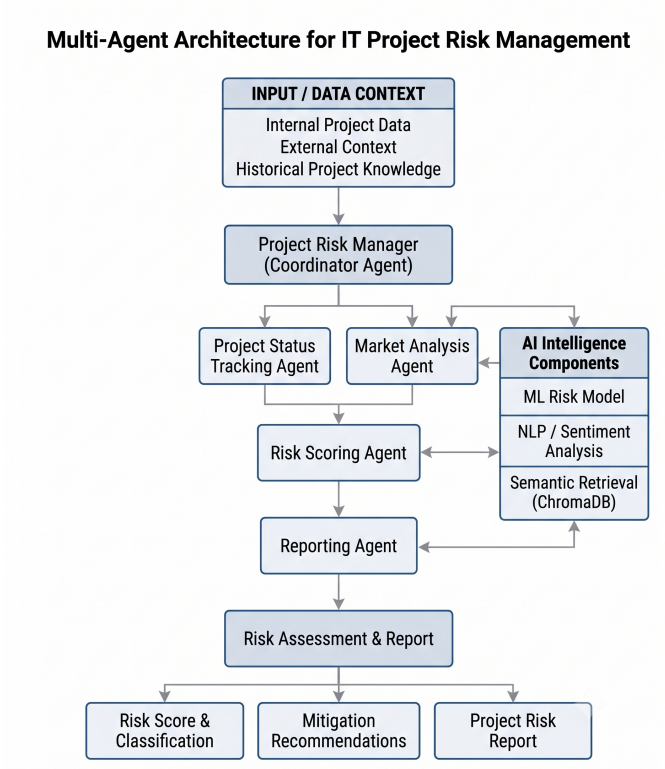



Fig. 2. Multi-agent workflow for coordinated IT project risk assessment

### A. Project Risk Manager

The Project Risk Manager is the coordinator. The Project Risk Manager receives the project analysis request. The Project Risk Manager assigns tasks to agents. The Project Risk Manager checks the outputs from those agents. The Project Risk Manager brings all the results together. The Project Risk Manager is the controller of the risk analysis workflow.

### B. Project Status Tracking Agent

The Project Status Tracking Agent looks at internal project health indicators. The Project Status Tracking Agent checks schedule progress. The Project Status Tracking Agent checks milestone completion. The Project Status Tracking Agent checks team performance. The Project Status Tracking Agent checks reported project issues. The Project Status Tracking Agent finds deviations from expected conditions. The Project Status Tracking Agent writes observations that help the overall risk assessment.

### C. Market Analysis Agent

The Market Analysis Agent looks at information that matters to the project. The Market Analysis Agent uses NLP processing and sentiment analysis. The Market Analysis Agent pulls signals from news. The Market Analysis Agent pulls signals from industry developments. The Market Analysis Agent pulls signals from external information. These signals give context that goes beyond the projects internal metrics.

### D. Risk Scoring Agent

The Risk Scoring Agent brings together all the outputs from the analysis process. The Risk Scoring Agent creates a risk assessment for the project. The Risk Scoring Agent uses a machine learning risk model. The Risk Scoring Agent uses the analytical signals. The Risk Scoring Agent gives the project a risk score. The Risk Scoring Agent classifies the project as On Track In Progress or At Risk.

### E. Reporting Agent

The Reporting Agent turns the outputs into a structured risk report. The Reporting Agent writes a summary of the identified risks. The Reporting Agent writes a summary of the supporting observations. The Reporting Agent writes a summary of the risk assessment. The Reporting Agent writes a summary of the recommended mitigation actions. The Reporting Agent makes the report available on the dashboard.

The overall interaction, between the agents follows the sequence:

Project Risk Manager → Project Status Tracking Agent / Market Analysis Agent → Risk Scoring Agent → Reporting Agent → Dashboard

## V. Experimental Setup

### A. Dataset and Preprocessing

We test our machine-learning component on the credit risk dataset that is part of our prototype environment. This dataset holds structured attributes about borrowers and loans. The field loan_status is the target we want to predict. Before we train we fill missing values with the median and turn categorical variables into numbers.

We split the dataset into training and testing parts in an 80:20 ratio using a fixed state so the split is reproducible. We apply the preprocessing steps to every model that we test.

### B. ML Models

We assess two classification methods:

- Logistic Regression – a common baseline classification model.

- Random Forest – a non-linear ensemble model that can capture relationships among many risk-related features.

For the Random Forest model we build 100 trees and set a fixed state. We train both Logistic Regression and Random Forest with the training and testing sets so that the comparison is fair.

### C. Evaluation Metrices

We judge the classification models by accuracy, precision, recall and F1-score. We also look at the confusion matrix to see how many predictions are right or wrong for each risk class.

In addition, to prediction quality we can evaluate the framework by measuring system-level aspects like risk-analysis latency and the execution time of each agent. These metrics let us see how the multi-agent workflow behaves in practice.

## VI. Results & Discussions

We have built a prototype that shows how structured risk analysis, NLP-based contextual analysis, semantic retrieval and multi-agent coordination can all work together inside one project-risk management workflow. The machine-learning part of the prototype helps classify risk with learning and the multi-agent layer of the prototype coordinates project status analysis, contextual analysis, risk scoring and report generation.

The prototype outputs three types of results: project-risk indicators, consolidated risk classifications and structured mitigation reports. The dashboard displays these results with important project-health data so that project stakeholders can see the factors that lead to the risk condition that the prototype reports.

The architecture of the prototype also keeps each component separate from the final risk assessment. This allows the outputs of the status-analysis component, the market-analysis component, the machine-learning component and the historical-retrieval component to be looked at on their own before the prototype brings them with the risk-scoring workflow.

However the prototype does not give any numbers for how the whole multi-agent system performs because a fully controlled test has not yet been done. In particular the prototype has not built an enough project-risk benchmark dataset or run many experiments to measure prediction accuracy, agent speed and how each component of the prototype helps. As a result the results that the prototype shows should be seen as a prototype-level check of the workflow not as a comparison with other risk-management methods.

The current implementation of the prototype gives a base, for quantitative tests that will use a special IT project-risk dataset and controlled experiments.

## VII. Limitations & Future Work

Our current prototype has a limits that point to ways we can grow it. First the system only uses project data that lives inside the prototype environment and not the steady data streams that come from project-management platforms. So we plan to connect to sources such, as Jira and GitHub so that the system can watch projects in time.

Second the current evaluation does not use an enough benchmark dataset to compare numbers fully. In work we will create or get a dedicated IT project-risk dataset and then test the framework with standard classification metrics, latency measurements and controlled experiments.

Third we will study how each analytical component helps by doing ablation studies. That way we can see the effect of project metrics NLP-based signals and historical knowledge retrieval on their own.

Finally we will grow the framework to monitor projects at once and to pull in richer historical knowledge. That will let the system learn from a set of projects that it has already looked at.

## VIII. Conclusion

We presented an AI-powered framework for IT project-risk management. This AI-powered framework for IT project-risk management fuses structured project analysis, machine learning NLP-based analysis, semantic retrieval and coordination among multiple agents. The AI-powered framework for IT project-risk management therefore shares its work load across agents and gathers all their results into a single unified risk assessment and a clear structured report.

The prototype shows how diverse project information can be folded into a common risk-management workflow. The prototype shows that of working in isolated techniques the AI-powered framework for IT project-risk management can process many different kinds of data together. Although full quantitative evaluation is still a task, this AI-powered framework for IT project-risk management gives a solid base, for building and testing an intelligent decision-support system that keeps checking IT project risk all the time.


## Acknowledgment

We would like to express our heartfelt gratitude to our mentor, Prof. Sourav Malakar, for his valuable guidance, continuous support, and constructive feedback throughout the development of this work. His encouragement and insights greatly helped us in shaping and improving our project. We are sincerely thankful for his time, guidance, and motivation throughout this work.


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